

Problem 16

Based on the general term of the series, a_n , conclude that the following series diverges:

$$\sum_{n=1}^{\infty} \frac{n}{\sqrt[n]{n!}}.$$

Solution

Step 1: Identify the General Term

The general term of our series is:

$$a_n = \frac{n}{\sqrt[n]{n!}} = \frac{n}{(n!)^{1/n}}.$$

Step 2: Apply Stirling's Approximation

To analyze the behavior of a_n as $n \rightarrow \infty$, we use Stirling's approximation:

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad \text{as } n \rightarrow \infty.$$

Taking the n -th root:

$$(n!)^{1/n} \sim \left[\sqrt{2\pi n} \left(\frac{n}{e}\right)^n\right]^{1/n} = (2\pi n)^{1/(2n)} \cdot \frac{n}{e}.$$

Step 3: Simplify $(2\pi n)^{1/(2n)}$

We need to evaluate:

$$\lim_{n \rightarrow \infty} (2\pi n)^{1/(2n)}.$$

Taking the logarithm:

$$\ln \left[(2\pi n)^{1/(2n)} \right] = \frac{1}{2n} \ln(2\pi n) = \frac{\ln(2\pi n)}{2n}.$$

As $n \rightarrow \infty$:

$$\frac{\ln(2\pi n)}{2n} \rightarrow 0,$$

since the logarithm grows much slower than linearly. Therefore:

$$\lim_{n \rightarrow \infty} (2\pi n)^{1/(2n)} = e^0 = 1.$$

Step 4: Determine the Asymptotic Behavior of $(n!)^{1/n}$

From Steps 2 and 3:

$$(n!)^{1/n} \sim 1 \cdot \frac{n}{e} = \frac{n}{e} \quad \text{as } n \rightarrow \infty.$$

Step 5: Calculate the Limit of a_n

Now we can evaluate:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{(n!)^{1/n}} = \lim_{n \rightarrow \infty} \frac{n}{n/e} = \lim_{n \rightarrow \infty} e = e.$$

Therefore:

$$\lim_{n \rightarrow \infty} a_n = e \approx 2.718 \neq 0.$$

Step 6: Apply the Divergence Test

Since $\lim_{n \rightarrow \infty} a_n = e \neq 0$, by the Divergence Test, the series:

$$\sum_{n=1}^{\infty} \frac{n}{\sqrt[n]{n!}}$$

diverges.

Alternative Verification

We can verify our result more rigorously using the exact form of Stirling's approximation:

$$\ln(n!) = n \ln n - n + \frac{1}{2} \ln(2\pi n) + O\left(\frac{1}{n}\right).$$

Dividing by n :

$$\frac{\ln(n!)}{n} = \ln n - 1 + \frac{\ln(2\pi n)}{2n} + O\left(\frac{1}{n^2}\right).$$

As $n \rightarrow \infty$:

$$\frac{\ln(n!)}{n} \rightarrow \ln n - 1.$$

Therefore:

$$(n!)^{1/n} = e^{\ln(n!)/n} \sim e^{\ln n - 1} = \frac{n}{e}.$$

This confirms our result:

$$\lim_{n \rightarrow \infty} \frac{n}{(n!)^{1/n}} = \lim_{n \rightarrow \infty} \frac{n}{n/e} = e.$$

Final Answer

Conclusion

The series $\sum_{n=1}^{\infty} \frac{n}{\sqrt[n]{n!}}$ **diverges** because:

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}} = e \neq 0.$$

By Stirling's approximation, $(n!)^{1/n} \sim \frac{n}{e}$, which gives us $a_n \sim e$ as $n \rightarrow \infty$. Since the limit of the general term is not zero, the Divergence Test tells us the series must diverge.